Alex Chen

· He/him · Avid hiker
· 34 years old · High School Counselor

Personality: Agreeableness

Maya Sharma

· She/her · Guitar player
· 17 years old · High School Junior

Personality: Conscientiousness

Scenario: Maya failed in the math exam and asks advice from Alex.
Alex's goal: Make Maya feel safe from this failure.
Maya's goal: Find relief from her anxiety.

Perception F_{per}

Understand the partner's previous-turn video

Input: Perceived Alex video from turn $t-1$

Turn t

Turn $t+1$

Extracted Cues

Speech (T_s^t): *'Maya, I know you feel bad right now ...'*

Emotion State (T_e^t): *'supportive ...'*

Facial Expression (T_d^t): *'gentle; firm eyes ...'*

Social Reasoning F_{rea}

Infer hidden mental status and choose the best response

Structured Observation (O_a^t) | Persona Profile (P_a) | Scenario & Social Goal (G_a)

Theory-of-mind Inference

Belief (β^t): *'Alex is not judging me ...'*

Desire (δ^t): *'Alex wants to make me feel safe ...'*

Intention (ι^t): *'Alex is trying to comfort me...'*

Emotion (ε^t): *'Alex seems supportive and concerned ...'*

Utterance Candidates Generator (C^t)

Ensemble Selection

Empathy Check (U_{emp}) | Strategic Check (U_{strat}) | Coherence Check (U_{coh})

Best Utterance (D_a^t): *'Thanks Alex. I've done my best ... '*

Feeling (E_a^t): I feel Relieved but still Anxious.

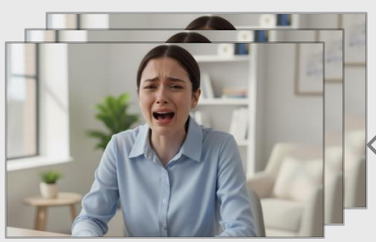
Turn t

Roles switch

Speaker (Maya)
⇄
Listener (Alex)

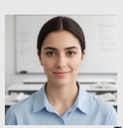
Expression F_{exp}

Render the selected response as a talking head video



Output: generated Maya video at turn t.

Video Diffusion Model (G_{video})



portrait image (I_{ref}^a) reference audio (a_a^t)

Turn t